



Assignment 2

CSE260: Digital Logic Design

Department of Computer Science and Engineering

Semester : Spring 26

Marks: 5

Graded (5 marks - 1 mark each)

1. **Simplify** the following Boolean expressions:
 - a. $(P + Q + R)(P + Q')(P' + R)$
 - b. $A'B'C + (A+B+C)' + A'B'C'$
 - c. $((AB)'.(C+D)')' + (A \square (B+C))$
2. Find the **complement** of the following Boolean expressions via duality:
 $A' + C(B + C')(A' + B')(A + C')$

3. From the truth table, find out the POS. No need to simplify the result.

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

4. Draw the following functions using **NOR gates only** (without simplification):

$$F(X, Y, Z) = (X' + YZ + XZ') + X'Y + (X+Z')$$

5. Find **SOP** for the following Boolean function: $F(P, Q, R, S) = P + QR'S$